

USHASWI KURMALA

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EDUCATION

Master of Science - Data Science, State University of New York at Buffalo (CGPA: 3.66/4) August 2023 - December 2024

Relevant Coursework: Machine Learning, Python programming, Data Model and Query Languages, Storytelling through Data Visualization, Statistical Analysis, Numerical Mathematics, Data Analysis

EXPERIENCE

Data Engineer

June 2021 - July 2023

NP Solutions

Hyderabad, India

- Designed scalable ETL pipelines with AWS Glue and PySpark to process 500TB of financial data, including stock market trends, historical prices, trading volumes, and sentiment datasets, Cutting down data processing time by 31% and enhancing reporting efficiency.
- Preprocessed large-scale stock market data using Python and SQL, resolving 94% of data anomalies and ensuring data integrity for investment analysis and forecasting market trends.
- Deployed machine learning algorithms such as Random Forest and Gradient Boosting within AWS SageMaker to predict stock price fluctuations and market volatility, achieving 84% prediction accuracy, and boosted data retrieval performance in AWS Redshift by 42%.
- Performed advanced statistical analysis on stock portfolios, volatility patterns, and trading signals, generating insights that drove strategy development and increased trading efficiency by 32%.
- Created dynamic Tableau dashboards to track stock market KPIs, including volatility indices, market sentiment, and trading volume, delivering real-time decision-making support and improving efficiency by 26%.
- Delivered comprehensive reports with strategic insights on market trends, translating complex analyses into clear action plans, minimizing risk exposure and maximizing returns on investments.

PROJECTS

Credit Risk and Loan Default Prediction Analysis - (Python, Machine Learning, Data Processing, Data Visualization)

- Utilized advanced machine learning techniques to predict loan defaults, revising risk assessment processes and decreasing default rate by 22%.
- Optimized data pipelines for faster processing, allowing credit risk team to make quicker, data-driven decisions on loan approvals and mitigating financial risk.

Insurance Claim Approval Analysis and Dashboard Development - (Data Analysis, EDA, Regression, Tableau)

- Built a fraud detection model analyzing 10,000+ insurance claims, identifying key patterns and reducing fraudulent claims by 17%.
- Boosted claim validation processes, decreasing approval times and optimizing operational efficiency by 18% through real-time insights and automated alerts.

Investment Portfolio Optimization and Rebalancing - (Python, SQL, ML, Data Analysis, Financial Modelling)

- Developed an automated investment portfolio optimization system using historical stock data, incorporating mean-variance optimization and modern portfolio theory to increase returns while minimizing risk.
- Implemented a real-time rebalancing system, boosting portfolio returns by 18% by continuously optimizing asset allocation based on market shifts.

SKILLS

Python (Pandas, NumPy, PyFinance), SQL, ETL, AWS, Airflow, Matplotlib, Seaborn, Hadoop, Spark, Power BI, Tableau, Risk Modeling, Portfolio Optimization, Financial Metrics Analysis, Statistical Analysis (VaR, regression), Machine Learning, Predictive Analytics (time series forecasting), Agile, Waterfall, CI/CD, Jenkins, GitHub,

CERTIFICATIONS

- PL-300 Microsoft Certified: Power BI Data Analyst Associate
- Foundations: Data, Data, everywhere - Google, Coursera
- IBM Data Analyst Professional Certificate